stockplotr: :CHEATSHEET

Create report-ready tables and figures for stock assessments.

The Basics

Convert output file to a standardized framework.

```
data <- convert_output(
  file = <FILE_PATH>,
  model = "MODEL ACRONYM",
  fleet_names = c("FLEET_NAMES"),
  save_dir = <FILE_PATH>
)
```

Optional

label	estimate	year	fleet	sex	...
spawning_biomass	2017090	2016	NA	NA	...
spawning_biomass	2286740	2017	NA	NA	
...	...	...	...	...	...

Create customizable plots.

```
plot_biomass(
  <DATA>,
  geom = <GEOM>,
  group = "<COLUMN_NAME>",
  facet = "<COLUMN_NAME>",
  ref_line = "msy",
  unit_label = "metric tons",
  module = "MODULE_NAME",
  scale_amount = 1,
  relative = FALSE,
  make_rda = FALSE,
  figures_dir = <FILE PATH>,
  interactive = TRUE,
  …
)
```

"line", "point" or "area"

Export object as an .rda

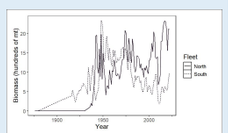
Additional ggplot2 arguments

Example

plot_biomass(example_data, unit_label = "mt", ref_line = "target", scale_amount = 100, module = "TIME_SERIES", make_rda = TRUE, figures_dir = getwd())

load(biomass_figure.rda)

rda$figure



rda$caption

> "Biomass (B) time series. The horizontal dashed line represents the limit reference point (msy mt)."

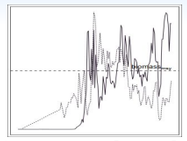
rda$alt_text

> "Line graph showing biomass time series. The x axis shows the year, which spans from 1874 to 2022. The y axis shows biomass in hundreds of mt, which spans from 54.2 to 426.49."

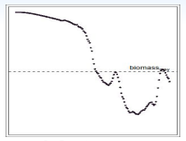
Figures

time series

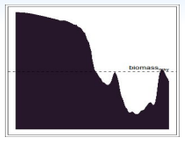
line



point



area



plot_spawning_biomass(dat, geom, …)
Line, point, or area over time with error where applicable.

plot_biomass(dat, …)
Total biomass of line, point, or area over time with error where applicable.

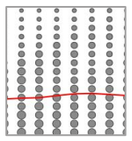
plot_recruitment(dat, …)
Observed and expected recruitment over time with error where applicable.

plot_spawn_recruitment(dat, …)
Stock recruitment relationship with fitted line.

plot_fishing_mortality(dat, …)
Line or points of fishing mortality over time.

age series

plot_abundance_at_age(dat, z, proportional, …)
Bubble plot depicting abundance at age over time with trend lines across the x and y axes.

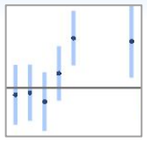


plot_biomass_at_age(dat, …)
Bubble plot representing biomass at age over time.

plot_catch_comp(dat, …)
Bubble plot representing catch at age over time with trend lines of top 5% strongest cohorts.

miscellaneous

plot_recruitment_deviations(dat, …)
Scatter plot of points with line segments depicting error.



plot_indices(dat, …)
Observed vs. estimated indices of abundance by fleet with error where applicable.

Tables

Tables follow the same format and conventions of figures with the exception of facetting.

table_landings(dat, …)
Total or indexed landings over time including error.

table_indices(dat, …)
Indices of abundance by fleet over time.

table_derived_quantities(dat, …)
Specified derived quantities for management purposes. Default includes: SBmsy, Ftarget, R0, steepness.

Theming

1. **Apply NOAA Fisheries themes to both figures and tables. Applies to objects made from `ggplot2`, `flextable`, and `gt`.**

   library(ggplot2)
   plot <- ggplot(cars) + geom_line(aes(x = speed, y = dist))
   add_theme(plot)

2. **Add theme onto `ggplot2` object using (+) operator**

   library(ggplot2)
   ggplot(cars) +
   geom_line(aes(x = speed, y = dist)) +
   theme_noaa()

Exporting

save_all_plots(dat, …)
Save each available plot and table in `stockplotr` as an rda in a folder called "figures".

html_all_figs_tables(figures_tables_dir)
Create an html file containing all plots and tables from `stockplotr` to view together. Requires a folder containing rda files made from either `save_all_plots()` or individual plots and tables.

Follow along with our planning and development on our GitHub project! Be on the lookout for future releases including more tables and figures!

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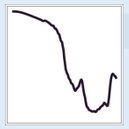
Figures

Plot Types

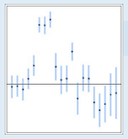
(All base plotting functions accept dat (data), x, y, group, facet, xlab, and ylab.)



plot_timeseries(dat, geom, …)

Line, point, or area over time with error where applicable.

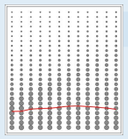
plot_timeseries(
dat = data, group = "fleet", facet = "area")



plot_error(dat, …)

Scatter plot of points with line segments depicting error.

plot_error(petrale, …)



plot_aa(dat, z, proportional …)

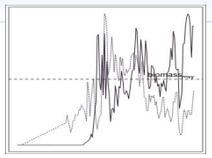
Bubble plot depicting estimates at age over time with trend lines across the x and y.

plot_aa(petrale, facet = "area", proportional = TRUE)

Automated

Standard XY Plots

time series



plot_spawning_biomass()
dat, geom, group, facet, ref_line, unit_label, scale_amount, relative

plot_biomass(dat, …)
plot_recruitment(dat, …)
plot_landings(dat, …)
plot_fishing_mortality(dat, …)

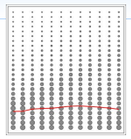
age series

plot_natural_mortality(dat, …)

miscellaneous

plot_spawn_recruitment(dat, …)

"At-age" Bubble Plots



plot_abundance_at_age(dat, z, proportional …)

Bubble plot depicting estimates at age over time with trend lines across the x and y.

plot_biomass_at_age(dat, …)

Bubble plot representing biomass by age over time.

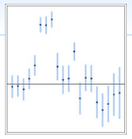
plot_biomass_at_age(petrale, facet = "sex")

plot_catch_comp(dat, …)

Bubble plot representing catch by age over time with trend lines of top 5% strongest cohorts.

plot_catch_comp(petrale, …)

Error



plot_recruitment_deviations(dat, …)

Scatter plot of points with line segments depicting error.

plot_recruitment_deviations(petrale, …)

Theming + Customization

```
a + theme_noaa(discrete = TRUE)
```

```
plot_spawning_biomass(
   example_data,
   group = "fleet") +
ggplot2::geom_hline(ggplot2::aes(
   yintercept = 1000))
```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Tables

Prepare data for table

prepare_table()

Filter data to be ready to input into a plotting or table function.

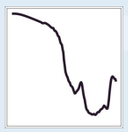
```
data <- stockplotr::prepare_table(
  dat = data,
  label_name = "quantity name to plot",
  geom = "line",
  era = NULL,
  group = "indexing column",
  facet = "column(s) for  facetting",
  interactive = TRUE
)
```

Optional: "point" or "area"

Year	Biomass (mt)	Catch (mt)	Fishing Mortality
2000	2017090	43092	*0.20*
2001	2286740	42095	*0.21*
2002	2190210	90283	0.43
...	...	...	...

Plot Types

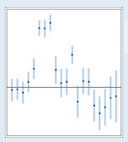
(All base plotting functions accept dat (data), x, y, group, facet, xlab, and ylab.)



plot_timeseries(dat, geom, …)

Line, point, or area over time with error where applicable.

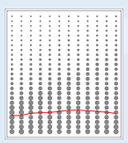
plot_timeseries(petrale, group = "fleet", facet = "area")



plot_error(dat, …)

Scatter plot of points with line segments depicting error.

plot_error(petrale, …)



plot_aa(dat, z, proportional …)

Bubble plot depicting estimates at age over time with trend lines across the x and y.

plot_aa(petrale, facet = "area", proportional = TRUE)

Automated Functions

Create tables – no work required.

table_spawning_biomass(geom = "line")
dat, geom, group, facet, ref_line, unit_label, scale_amount, relative

table_spawning_biomass(geom = "point**)**
dat, geom, group, facet, ref_line, unit_label, scale_amount, relative

table_spawning_biomass(geom = "area**)**
dat, group, ref_line, unit_label, scale_amount, relative

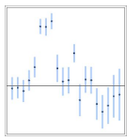
table_biomass(dat, …**)**

table_recruitment(dat, …**)**

table_spawn_recruitment(dat, …**)**

table_landings(dat, …**)**

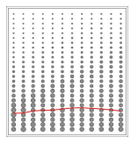
table_fishing_mortality(dat, …**)**



plot_recruitment_deviations(dat, …)

Scatter plot of points with line segments depicting error.

plot_recruitment_deviations(petrale, …)



plot_abundance_at_age(dat, z, proportional …)

Bubble plot depicting estimates at age over time with trend lines across the x and y.

plot_biomass_at_age(dat, …)

Bubble plot representing biomass by age over time.

plot_biomass_at_age(petrale, facet = "sex")

plot_catch_comp(dat, …)

Bubble plot representing catch by age over time with trend lines of top 5% strongest cohorts.

plot_catch_comp(petrale, …)

Exporting

1. **Execute all functions at once**
2. **Render and compile all functions into a readable html for interpretation**

save_all_plots(dat, …)

Save each available plot and table in `stockplotr` as an rda in a folder called "figures".

save_all_plots(petrale, …)

html_all_figs_tables(dat, …)

Create an html file containing all plots and tables from `stockplotr` to view together.

save_all_plots(petrale, …)

Theming

Create and automate alternative text & captions. Any plot in `stockplotr` exported as an **.rda will contain alternative text and a caption** customized to the data using **key quantities**. A csv file is generated containing captions and alternative text associated with every plot and table that could be produced.

label	type	caption	alt_text
spawning_biomass	figures	Model-estimated spawning biomass…	*Line graph showing …*
catch	table	Observed catch over time…	*NA*
recruitment	figure	Estimated recruitment by …	Line graph showing…
...	...	...	...

.RDA Package & Key Quantities

- *make_rda = TRUE* → **produce .rda file** containing (1) *figure/table*, (2) *caption*, (3) *alternative text*
- Quantities extracted from data file
- Replace **placeholders** in caption and alternative text to maintain standardization while customizing reported values

```
plot_spawning_biomass(
  dat = petrale,
  make_rda = TRUE
 )
load("figures/spawning.biomass_figure.rda")
rda$alt_text
rda$cap
```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asar’s former convert_output() section in cheatsheet

⬠→△ Convert model output

Convert output file to a standardized framework

```
output <- asar::convert_output(
  output_file = "Report .sso”,
  model = "ss3")
```

SS3 files supported

```
output <- asar::convert_output(
  output_file = "Report .rdat”,
  model = "bam")
```

BAM files supported

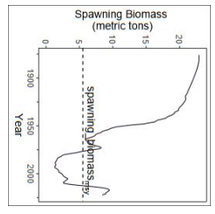
```
output <- asar::convert_output(
  output_file = estimates,
  model = "fims")
```

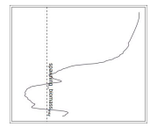
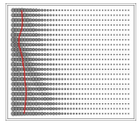
FIMS files supported *(tibble in R environment* (shown here) *OR .rds file)*

Standardized framework format *(dataframe in .rda file)*

label	estimate	year	fleet	sex	...
spawning_biomass	2017090	2016	*NA*	*NA*	...
spawning_biomass	2286740	2017	*NA*	*NA*	
...	...	...	...	...	...

ℹ️ See “Standardizing Assessment Model Output” vignette


Spawning Biomass
(metric tons)
20
15
10
5
0
spawning biomass$_{MSY}$
1900
1950
2000
Year


spawning biomass$_{MSY}$